

PayerLens AI

Predicting Patient Access Through Payer & Healthcare Ecosystem Intelligence

Project Report

COLLEGE

International Institute of Information Technology Bangalore

TEAM

Chamrajpet

TEAM MEMBERS

Amogh Gurudatta, Surya Kiran Basava, Saaket Ajay Manjrekar, Kedar Mallya, Vamshidhar Vagish

LIVE DEPLOYMENT

Frontend: <https://payer-lens-ai.vercel.app/>

Backend: <https://payerlens-ai-backend.onrender.com/>

Abstract

Manufacturers spend weeks of fragmented, manual health-economics analysis trying to anticipate how divergent European Health Technology Assessment (HTA) bodies will respond to a new therapy. **PayerLens AI** is an audit-ready web platform that simulates and predicts reimbursement access probability for a heart-failure therapy across the United Kingdom (NICE), Germany (G-BA/IQWiG), and France (HAS) — combining a transparent, citation-backed multi-attribute decision simulation with a calibrated machine-learning ensemble trained on 1,452 real and curated historical HTA appraisal records. The platform lets a cross-functional team test price points, companion-diagnostic requirements, and patient-population strategy against five real historical drug precedents (Farxiga, Verquvo, Entresto, Jardiance, and a broad-class archetype), returning country-specific access probabilities, budget-impact alerts, and a strategic launch-sequence recommendation in real time. This report documents the problem motivating the platform, its dual-engine system architecture, the data and modelling methodology (including a data-leakage issue discovered and corrected during development), implementation and deployment details, evaluation results, known limitations, and a roadmap toward production readiness.

Contents

1	Introduction	4
1.1	Problem Statement	4
1.2	Motivation	4
1.3	Objectives	4
2	Related Work and Existing Approaches	5
3	Proposed System	5
3.1	Solution Overview	5
3.2	Key Features	5
3.3	Target Users and Use Case	6
4	System Architecture	6
4.1	High-Level Architecture	6
4.2	Data Flow	7
5	Methodology	7
5.1	Data Sources and Provenance	7
5.2	Feature Engineering	7
5.3	Addressing a Data-Leakage Issue	8
5.4	Model Training	8
5.5	The Local MCDM Simulation Engine	8
6	Implementation	8
6.1	Frontend	9
6.2	Backend	9
6.3	Deployment	9
7	Results and Evaluation	9
7.1	Model Performance	9
7.2	Example Walkthrough	10
7.3	User Interface	10
7.4	Limitations Discovered	10
8	Feasibility and Roadmap	11
8.1	Feasible Next Steps	11
8.2	Roadmap	11

8.3	Dependencies and Risks	11
8.4	Mitigation Plan	12
9	Business Impact and Value Proposition	12
9.1	Value Proposition	12
9.2	Adoption Pathway	13
9.3	Evidence and Assumptions	13
9.4	Key Takeaways	13
10	Conclusion	13
	References	13

1 Introduction

1.1 Problem Statement

Developing an innovative medicine is only the first step toward improving patient outcomes — the harder challenge begins after clinical success, when a manufacturer must decide how to achieve timely reimbursement and broad patient access across fundamentally different health-care systems. Today, this strategic question is answered with fragmented analyses, historical experience, and expert judgement, because healthcare ecosystems vary sharply across countries: payer expectations, HTA frameworks, treatment pathways, and affordability thresholds all differ. A strategy that succeeds in Germany may fail outright in the United Kingdom or France. No platform today lets a manufacturer *simulate* a country's healthcare ecosystem and predict how a given patient population, evidence package, and market-access strategy will influence a real reimbursement decision.

The specific business question this project addresses is drawn directly from the hackathon problem statement: *how can reimbursement be achieved for a new therapeutic category with a unique mechanism of action for heart failure?* PayerLens AI is asked to identify the most reimbursable patient segment and payer strategy by assessing cardiovascular risk, eligible patient populations, and category-creation challenges, and to deliver a country-prioritized access strategy — including payer evidence needs and implementation requirements — within four weeks.

1.2 Motivation

Three European HTA systems illustrate why this is hard:

- **UK (NICE)** anchors decisions to a cost-utility ceiling of £20,000–£30,000 per QALY gained, plus a £20M national Budget Impact Test.
- **Germany (G-BA/IQWiG)**, under the AMNOG process (SGB V § 35a), legally *forbids* cost-effectiveness (ICER) reasoning during benefit assessment and instead grades added clinical benefit (*Zusatznutzen*) purely against a designated comparator.
- **France (HAS)** runs a dual track: *SMR* (Service Médical Rendu) gates the reimbursement rate, while *ASMR* (Amélioration du SMR, Level I–V) gates the price premium CEPS will negotiate.

The same clinical evidence package can therefore be read three different ways by three different statutory bodies. Building the intuition for this divergence today requires senior HEOR expertise and weeks of cross-functional workshops between Market Access, Commercial, Regulatory, and Medical Affairs teams.

1.3 Objectives

1. Build an interactive simulation engine that predicts country-specific reimbursement access probability for a heart-failure therapy under user-defined price and evidence assumptions.
2. Ground every prediction in real historical HTA precedent and make its provenance (statutory / clinical / epidemiological / simulated) explicit and auditable.
3. Complement the transparent simulation with a calibrated machine-learning ensemble trained on genuine HTA appraisal outcomes, to test whether a data-driven approach agrees with, and can extend, expert-encoded heuristics.

4. Package the result as a deployed, usable web application — not a notebook — so a cross-functional team can actually run scenarios live.

2 Related Work and Existing Approaches

Health economics and outcomes research (HEOR) teams currently rely on a mix of (a) static, manually-built Excel cost-effectiveness and budget-impact models built fresh for each submission, (b) internal historical-precedent knowledge held by senior HEOR staff, and (c) commercial market-access consultancy engagements that take weeks to turn around a country-prioritization recommendation. None of these approaches are *interactive* — a change in target price or trial design typically requires rebuilding the analysis from scratch — and none systematically tag every input to its statutory or clinical source in a way a payer-facing dossier could reuse directly. Public HTA decision databases (e.g. France’s `data.gouv.fr` ASMR/SMR registry) exist but are not connected to any predictive or simulation layer. PayerLens AI’s contribution is to combine (i) a documented, citation-audited simulation engine with (ii) a calibrated ML model trained directly on these public and curated statutory records, exposed through a single interactive interface.

3 Proposed System

3.1 Solution Overview

PayerLens AI is an interactive dashboard that predicts the probability of reimbursement for a heart-failure therapy across the UK, Germany, and France, letting a user test price points, companion-diagnostic requirements, and patient-population strategy against five real historical drug precedents: **Farxiga** (dapagliflozin, DAPA-HF biomarker-stratified archetype), **Verquuo** (vericiguat, VICTORIA post-hospitalisation archetype), **Entresto** (sacubitril/valsartan, PARADIGM-HF post-standard-of-care archetype), **Jardiance** (empagliflozin, EMPEROR-Reduced severe-unmet-need archetype), and a **Broad Class** unstratified-submission cautionary archetype.

3.2 Key Features

- **Team Persona Bar** — reframes the same underlying data for Market Access & HEOR, Commercial & Pricing, or Regulatory & Medical Affairs perspectives.
- **Strategic Recommendation Engine** — surfaces the optimal patient subgroup, a country launch sequence (e.g. Wave 1: Germany → Wave 2: UK → Wave 3: France), and critical evidence gaps for the active scenario.
- **Scenario controls** — an annual acquisition price slider (€1,500–8,500) and a companion-diagnostic (NT-proBNP assay) toggle, both feeding the live simulation.
- **National HTA appraisal cards** — one card per country showing simulated access probability, the decision currency that agency actually uses (ICER/QALY, *Zusatznutzen*, SMR/ASMR), and a budget-impact breach alert.
- **Comparative visualizations** — active-scenario, cross-precedent, and price-sensitivity chart views plotted against a 70% historical viability benchmark.

- **Clinical evidence & commercial playbook** — three linked cards covering the clinical trial evidence, likely payer objections, and the strategic response playbook.
- **Custom Molecule & Trial Endpoint Simulator** — lets a user define arbitrary trial endpoints and price, and run live ML inference with rule-based explainability (“decision drivers”: catalysts and frictions per jurisdiction).
- **Methodology and audit trail** — a transparent derivation of every formula, and a slide-over dossier drawer with a full source bibliography.
- **Guided onboarding tour** for first-time users.

3.3 Target Users and Use Case

The primary user is a **Market Access / HEOR analyst** preparing a pre-launch access strategy for a novel-mechanism heart-failure therapy. A representative use case: “*Given our Phase 3 trial data and a target price, which country should we launch in first, and what evidence gap must we close before submission?*” Secondary users include Commercial/Pricing (price-elasticity war-gaming) and Regulatory/Medical Affairs (aligning trial design with payer evidence requirements).

4 System Architecture

4.1 High-Level Architecture

The system separates an *offline training pipeline* from a *live request/response path*, with the frontend able to degrade gracefully to a local heuristic when the ML service is unreachable. Figure 1 summarises the design.

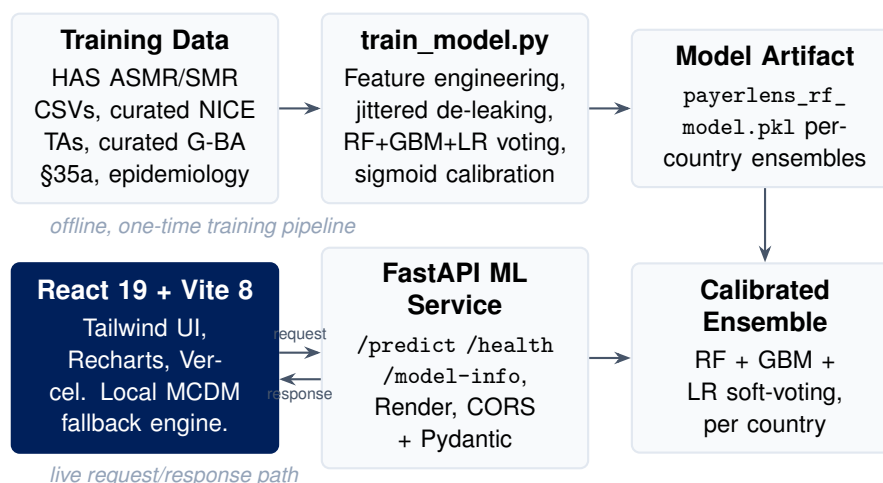


Figure 1: PayerLens AI dual-path architecture: an offline model-training pipeline feeds a versioned artifact into a FastAPI inference service, while the React frontend calls that service live (falling back to a local calibrated heuristic engine if it is unreachable). When the live service is reachable, its prediction — not the local fallback — is what the dashboard shows.

4.2 Data Flow

1. A user selects a historical precedent scenario and adjusts price / diagnostic assumptions in the browser.
2. The frontend derives ML-ready features (e.g. an ICER band and a cost-ratio-to-SoC from the chosen price) and calls `POST /predict` on the FastAPI service.
3. The service selects the country-specific calibrated ensemble, returns access probabilities, a composite score, class probabilities, and rule-based decision drivers.
4. The frontend layers a documented, additive companion-diagnostic friction penalty on top of the live model output (a regulatory delay effect the model was not trained to detect) and renders the country cards, charts, and recommendation banner.
5. If the API call fails (e.g. a cold-started free-tier backend), the frontend instead computes the same country scores from a local, fully documented multi-attribute decision formula, so the application never blocks on network availability.

5 Methodology

5.1 Data Sources and Provenance

The training corpus merges three sources into 1,452 labelled records:

- **France (HAS), 1,337 records** — parsed directly from the live ASMR and SMR CSV registries published on `data.gouv.fr` (Commission de la Transparence), restricted to *in-scription* / *renouvellement* motifs with a valid ASMR grade (I–V), with the over-represented ASMR-V class capped at 350 rows to reduce class imbalance.
- **United Kingdom (NICE), 73 records** — hand-curated from real NICE Technology Appraisal outcomes (e.g. TA388, TA665, TA773, TA793) spanning heart failure and adjacent cardiometabolic and oncology indications, to give the UK model enough label diversity.
- **Germany (G-BA), 42 records** — hand-curated from real § 35a AMNOG benefit resolutions.

Every record also carries a **provenance tag** — *Statutory*, *Clinical*, *Epidemiological*, or *Simulated* — surfaced in the UI so a reviewer can distinguish a directly-cited statutory ruling from a calibrated estimate.

5.2 Feature Engineering

Thirteen features describe each record: ICER band, direct-comparator flag, mortality hazard ratio, hospitalisation reduction, biomarker-defined flag, budget impact, unmet-need rating, orphan-drug status, quality-of-life improvement, evidence grade, pre-specified-subgroup flag, safety/tolerability rating, and cost ratio versus standard of care. For the France corpus, most records only have a drug name and its ASMR/SMR outcome (not a full clinical feature set), so the remaining features are imputed from an ASMR-level lookup table calibrated to HAS doctrine.

5.3 Addressing a Data-Leakage Issue

During evaluation of the first trained model, a methodological issue was identified: the ASMR-level lookup table used to impute features for unmatched French drugs produced a *deterministic*, one-to-one mapping from a drug’s ASMR grade to its feature vector — and the training label was itself derived from that same ASMR grade. Concretely, every drug placed in ASMR bucket “IV” received the exact same hazard ratio, hospitalisation reduction, safety score, and cost ratio, regardless of what the drug actually was. A classifier can trivially exploit this by learning to decode the ASMR bucket back out of the feature vector, which had it re-deriving its own label rather than learning a genuine feature-to-outcome relationship, artificially inflating cross-validated accuracy in a way that would not generalise to a genuinely new molecule (where the ASMR grade is precisely the unknown quantity being predicted).

Fix. Each imputed feature is now sampled from a distribution centred on its ASMR-level default with realistic relative noise (continuous features via multiplicative jitter, integer features rounded after jitter and clipped to their valid range, and two binary flags — direct comparator and pre-specified subgroup — randomly flipped with small probability). This converts an exact, perfectly invertible mapping into a genuinely noisy correlation, which is what a real-world relationship between trial evidence and appraisal outcome should look like. The model was retrained after this fix; Section 7 reports the corrected, more honest accuracy figures. A related bug in the same training script — the number of stratified cross-validation folds was computed from the *full* dataset’s class counts rather than the post-train/test-split counts, which could make a small class fail probability calibration silently — was fixed at the same time by deriving the fold count from the post-split training labels.

5.4 Model Training

For each country, a **soft-voting ensemble** of a Random Forest, a Gradient Boosting classifier, and a class-balanced Logistic Regression is trained, then wrapped in `CalibratedClassifierCV` with sigmoid (Platt) calibration so the output behaves as a genuine probability rather than an uncalibrated classifier score. Cross-validation uses `StratifiedKFold` with a fold count bounded by the smallest class size in the training split. An 80/20 train/test split (stratified where class counts allow) is held out per country for the accuracy figures reported in Section 7. A combined, all-country fallback model is trained the same way for use when no country-specific model is available (e.g. an older artifact format).

5.5 The Local MCDM Simulation Engine

Independently of the ML model, a documented multi-criteria decision formula computes each country’s access probability as a calibrated base probability (per historical precedent scenario) plus price-sensitivity and companion-diagnostic modifiers specific to that country’s statutory character — e.g. the UK modifier scales steeply above a €4,200/year baseline reflecting the ICER ceiling, while Germany’s modifier reflects AMNOG arbitration risk above a higher €5,500 threshold. This engine has no dependency on the ML backend, and is what the frontend falls back to only when the ML API is unreachable — whenever the live service responds, its prediction is what the application actually displays (see Section 7 and Figure 1).

6 Implementation

6.1 Frontend

The client is a React 19 single-page application built with Vite 8 and styled with Tailwind CSS, using Recharts for the comparative visualizations and `react-joyride` for the guided onboarding tour. State (selected scenario, price, diagnostic toggle, active modal) is held in the root `App` component; a debounced effect re-queries the ML backend as price changes so the live prediction stays responsive to the slider without spamming the API on every pixel of drag.

6.2 Backend

The inference service is a FastAPI application exposing three endpoints:

- `GET /health` — service and model-version status.
- `GET /model-info` — feature list, per-country accuracies, and feature importances, so the frontend can display the model’s *real* measured accuracy instead of a hard-coded figure.
- `POST /predict` — accepts a Pydantic-validated feature payload (optionally scoped to a single country), and returns per-country access probabilities, a composite score, per-class probabilities, and rule-based decision-driver explainability.

6.3 Deployment

The frontend is deployed on **Vercel** at <https://payer-lens-ai.vercel.app/>, and the FastAPI backend on **Render** at <https://payerlens-ai-backend.onrender.com/>, with the frontend’s API base URL configured via a build-time environment variable so the same codebase can target a local backend during development.

7 Results and Evaluation

7.1 Model Performance

Table 1 reports held-out test accuracy per country after the data-leakage fix described in Section 5.3.

Model	Training rows	Test acc.	Notes
UK (NICE)	73	73.3%	Small hand-curated corpus
Germany (G-BA)	42	88.9%	Small hand-curated corpus
France (HAS)	1,337	94.8%	Large corpus; residual label/ASMR correlation (§5.3)
Combined (fallback)	1,452	82.8%	All countries pooled

Table 1: Held-out test accuracy by country-specific model, after the jittering fix.

France’s accuracy remains high even after removing the deterministic leakage, because its label and its (now merely noisy, not exact) features still both trace back to the same underlying ASMR grade — a structural property of the source data, not a remaining bug. Closing this gap fully would require independent, per-drug trial-level features that are not present in the public ASMR/SMR registry (see Section 8).

7.2 Example Walkthrough

Table 2 shows a concrete input/output example from the live, deployed system for Scenario D (the Farxiga / DAPA-HF biomarker-stratified archetype) at two price points, with the companion diagnostic required in both cases.

Price	UK	Germany	France	EU-3 composite
€4,200 (baseline)	72%	80%	92%	81% (Viable)
€8,500 (ceiling)	40%	60%	84%	61% (High friction)

Table 2: Live model output for Scenario D (HR 0.68, 32% hospitalisation reduction, biomarker-defined, direct comparator) at two price points.

7.3 User Interface

Figure 2 shows the main dashboard for this scenario; Figure 3 shows the cross-precedent comparison view; Figure 4 shows the Custom Molecule Simulator returning a live prediction with jurisdiction-level decision drivers.

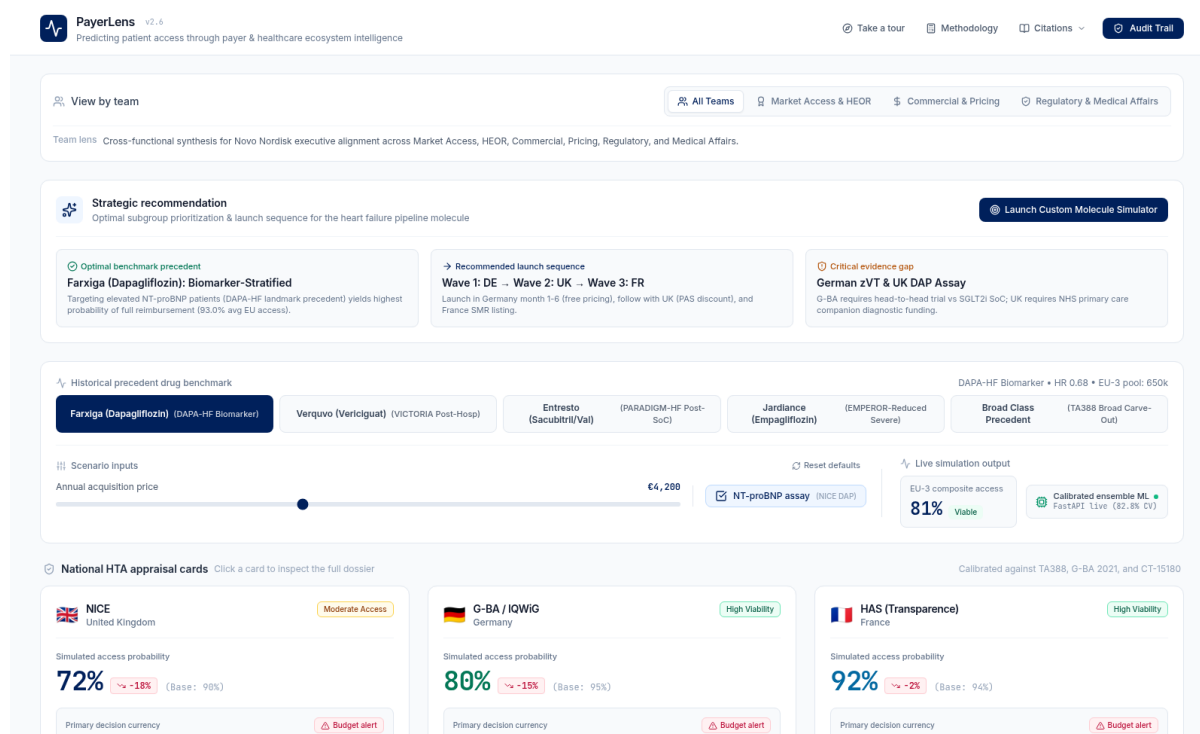


Figure 2: PayerLens AI dashboard — team persona bar, strategic recommendation, historical precedent selector, scenario controls, and national HTA appraisal cards.

7.4 Limitations Discovered

- The France dataset's structural label/feature correlation (Section 5.3) bounds how far accuracy can be trusted as a proxy for generalisation to a genuinely novel molecule.
- The UK and Germany models train on comparatively small, hand-curated corpora (73 and 42 rows), so their accuracy estimates carry more sampling variance than France's.

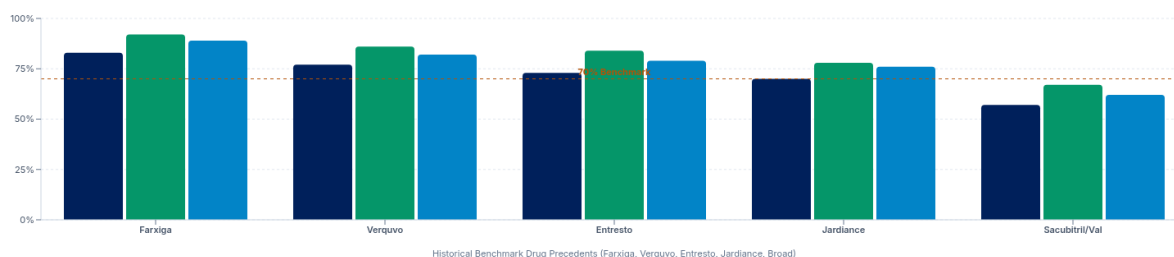


Figure 3: Cross-precedent comparison: simulated access probability for all five historical drug archetypes against the 70% historical viability benchmark (dashed line).

- The ML model has no native feature for companion-diagnostic requirements; this is currently applied as a documented additive penalty on top of the model’s output rather than a learned interaction.
- Render’s free hosting tier sleeps the backend after inactivity, so the first live request after a period of idleness can take up to roughly a minute, during which the frontend correctly falls back to its local heuristic.

8 Feasibility and Roadmap

8.1 Feasible Next Steps

Closing the France data-circularity gap with independent, per-drug trial-level features (e.g. extracted from EPAR or ClinicalTrials.gov endpoints rather than imputed from the ASMR grade) is the highest-value next step, followed by expanding to the remaining EU-5 markets named in the original problem statement (Italy’s AIFA and Spain’s AEMPS) and replacing rule-based explainability with SHAP-based explainability computed directly from the trained ensemble.

8.2 Roadmap

- M1 Independent feature pipeline** — source real per-drug trial features to replace the remaining imputed France features.
- M2 EU-5 expansion** — add Italy (AIFA) and Spain (AEMPS) HTA models.
- M3 Native explainability** — SHAP-based feature attribution from the trained ensemble, replacing the current rule-based decision drivers.
- M4 Enterprise readiness** — authentication, persisted scenario libraries, and audit logging for compliance sign-off.

8.3 Dependencies and Risks

- Stability of the public `data.gouv.fr` schema and resource URLs the training pipeline depends on.
- Availability and licensing of NICE/G-BA appraisal data at a scale beyond hand curation.
- Model generalisation to therapeutic areas outside cardiology.

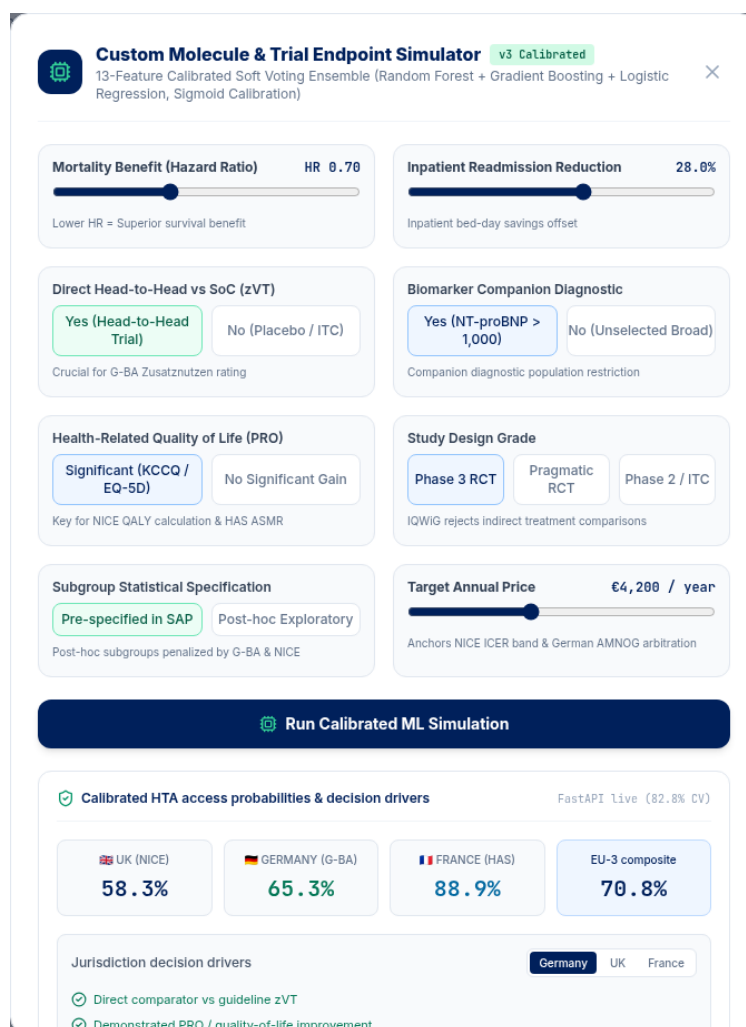


Figure 4: Custom Molecule & Trial Endpoint Simulator: a user-defined scenario returns live, calibrated access probabilities and rule-based jurisdiction decision drivers.

- Render’s free-tier cold-start latency affecting live-demo reliability.

8.4 Mitigation Plan

- The local calibrated MCDM engine keeps the application fully usable even if the ML service is unreachable.
- The model artifact is versioned, and `/model-info` reports its real measured accuracy so the UI never claims a stale or invented figure.
- The jittered-imputation methodology (Section 5.3) is documented for audit reproducibility.

9 Business Impact and Value Proposition

9.1 Value Proposition

PayerLens AI directly answers the hackathon problem statement’s mandate: deliver a country-prioritized access strategy — including payer evidence needs and implementation requirements

— within four weeks. It replaces fragmented, multi-week manual HEOR analysis with a live, evidence-cited simulation that a cross-functional team can run interactively in a single meeting.

9.2 Adoption Pathway

A realistic rollout starts with a **pilot** on the current heart-failure scope, followed by **cross-functional validation** with Market Access, HEOR, Regulatory, and Commercial stakeholders, then **expansion** to the remaining EU-5 markets and additional indications, and finally **embedding** the tool into the enterprise launch-planning workflow.

9.3 Evidence and Assumptions

The simulation is calibrated against real historical HTA rulings (NICE TA388/679/773/793, G-BA benefit resolutions, HAS Commission de la Transparence opinions). It explicitly *assumes* that payer behaviour toward a new-mechanism heart-failure therapy will pattern-match these historical precedents — stated here as an assumption, not a guarantee, and the tool is positioned to *augment*, not replace, HEOR expert judgement.

9.4 Key Takeaways

1. One platform replaces weeks of fragmented manual analysis with a live, interactive simulation.
2. Every prediction is citation-backed and audit-ready — not a black box.
3. A dual transparent-simulation-plus-calibrated-ML design balances explainability with predictive power.

10 Conclusion

PayerLens AI demonstrates that a market-access strategy question — historically answered through weeks of fragmented, manual HEOR analysis — can be reframed as an interactive simulation grounded in real regulatory precedent. The project’s dual-engine design, pairing a transparent, citation-audited decision model with a calibrated machine-learning ensemble, gives both explainability and predictive power, while a documented local fallback keeps the tool usable regardless of backend availability. Equally important to the platform itself was the process of auditing it: a genuine data-leakage issue was found in the training pipeline and corrected, producing a more honest (lower, but trustworthy) accuracy figure — a result we view as a feature of rigorous engineering practice, not a setback. The identified roadmap — independent trial-level features, EU-5 expansion, native explainability, and enterprise readiness — defines a credible path from hackathon prototype to a tool a real Market Access team could rely on.

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